UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 001 · Origin of electrode potential, cell potential · compiled 17 October 2026

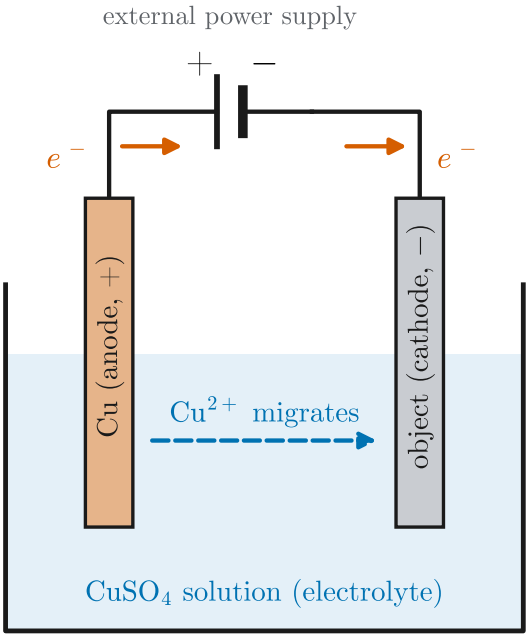
ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·10	Deck 001, slide 10 (rendered .pptx)
T001 00:08:22	AV-summary transcript 001, segment start
ESA Jan-25 s2 2b	Past paper: Jan 2025 set 2, question 2.b
ESA Jul-23a 4d	Past paper: Jul 2023 (first UE22 paper), question 4.d

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Colours follow the Okabe–Ito order. G2.1, G2.2 and G5.1 redraw slides 001·6, 001·7 and 001·10 (no screenshots). G1.1, G3.1, G3.2, G4.1 and G5.2 are schematics added for clarity (the deck has no cell diagram); they are not to scale.

1 Electrochemistry and the two types of cell



oxidation at anode: $\mathrm{Cu} \rightarrow \mathrm{Cu}^{2+} + 2\mathrm{e}^-$ reduction at cathode: $\mathrm{Cu}^{2+} + 2\mathrm{e}^- \rightarrow \mathrm{Cu}$

non-spontaneous reaction ($\Delta G > 0$) driven by the external supply

Figure G1.1. An electrolytic cell (copper electroplating). ESA ★

ADDED FOR CLARITY S001·3 · T001 00:02:50

How to read it. The external supply pushes electrons into the object (cathode, −), where Cu^{2+} is reduced and plated; the copper anode (+) is oxidised and replaces the ions. Compare with the galvanic Daniell cell (G3.1): there the reaction itself drives the electrons and the anode is negative.

2 Electrode potential and its origin: the Helmholtz double layer

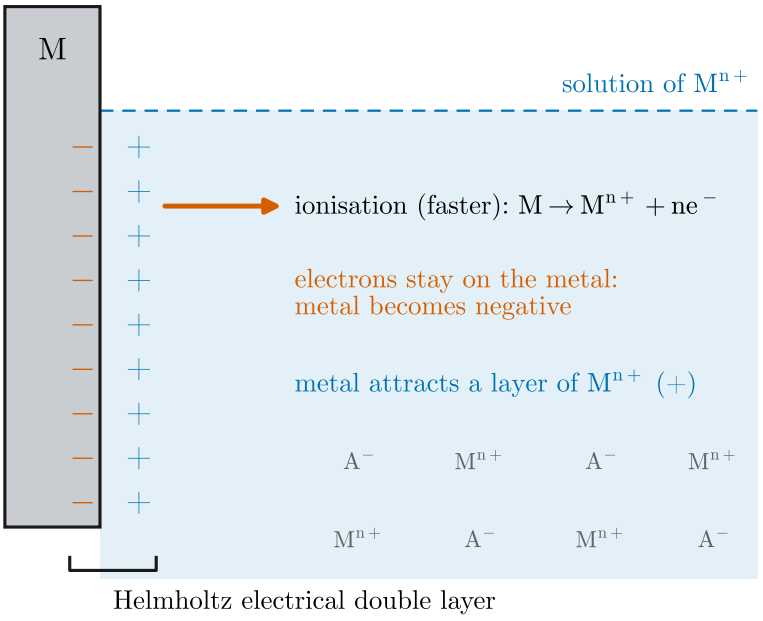


Figure G2.1. Case I: ionisation faster than deposition. ESA ★★★

S001·6 · T001 00:07:29 · ESA Jan-25 s2 2b

How to read it. Redrawn from slide 001·6. Draw the metal rod, a column of − charges on its surface, a parallel column of + ions in the solution, and label the pair "Helmholtz electrical double layer". Write $M \rightleftharpoons M^{n+} + ne^-$ beside it.

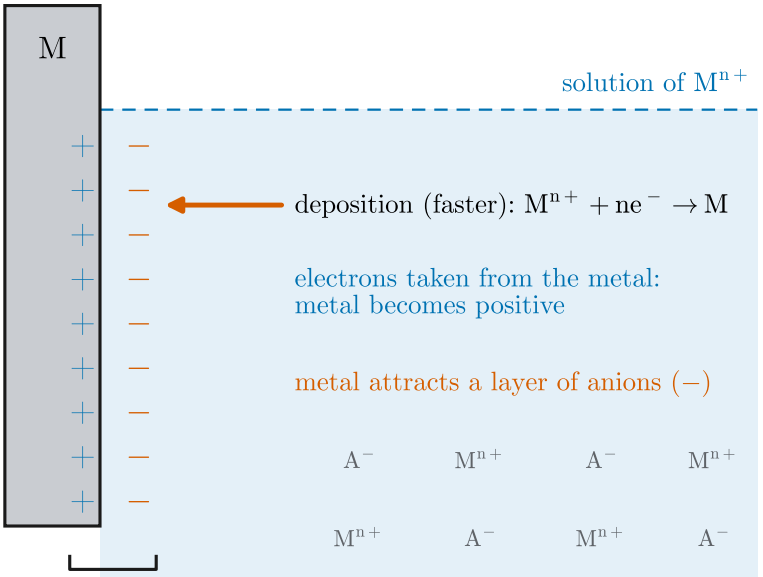


Figure G2.2. Case II: deposition faster than ionisation. ESA ★★

S001·7 · T001 00:08:40

How to read it. Redrawn from slide 001·7: the mirror image of G2.1. The rod carries + charges and attracts a layer of negative ions; write $\mathbf{M^{n+} + ne^- \rightleftharpoons M}$. The bulk anions $\mathbf{A^-}$ are added for clarity.

3 Standard electrode potential, electrochemical cell and cell notation

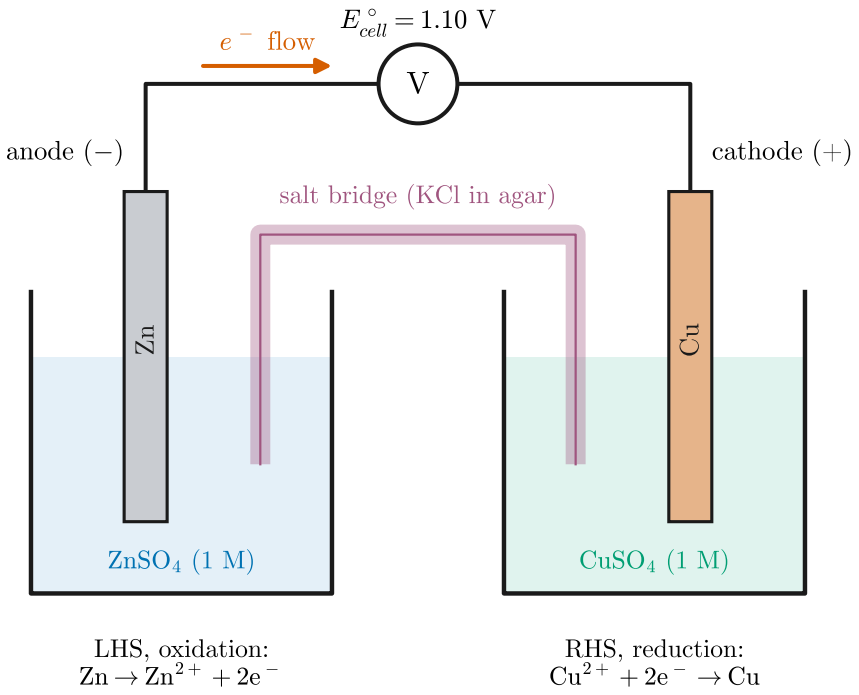


Figure G3.1. The Daniell cell. ESA ★★

ADDED FOR CLARITY S001·8 · T001 00:12:38

How to read it. Zinc in $\mathbf{ZnSO_4}$ (anode, −, left) and copper in $\mathbf{CuSO_4}$ (cathode, +, right), joined by a salt bridge between the solutions. Electrons flow in the external wire from Zn to Cu; the voltmeter reads $\boldsymbol{E^\circ_{cell}} = \mathbf{1.10}$ V at 1 M. The slide gives only the notation; the diagram is added. Schematic, not to scale.

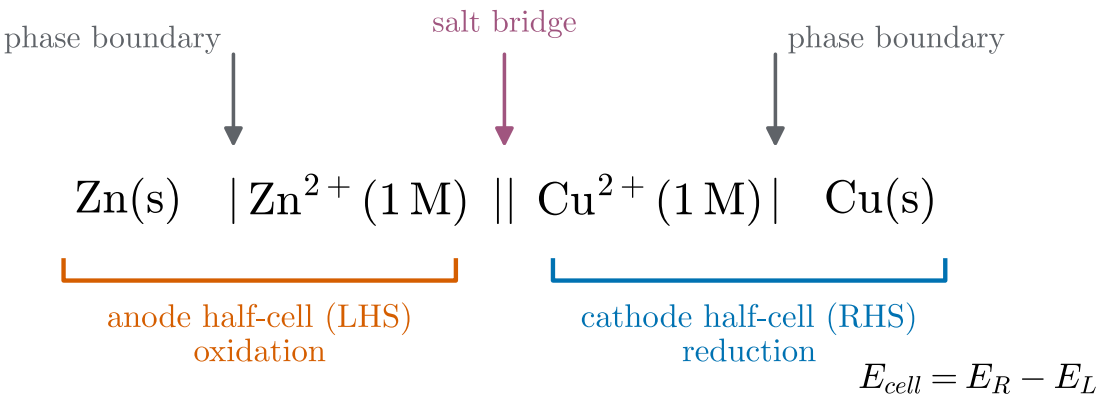


Figure G3.2. Reading the cell notation. ESA ★★★

S001·8 · T001 00:13:13

How to read it. Left half = anode (oxidation), right half = cathode (reduction). Each | is a phase boundary (solid metal | solution); || is the salt bridge, placed between the two solutions. E_{cell} = right minus left.

4 Cell potential, standard cell potential and free energy

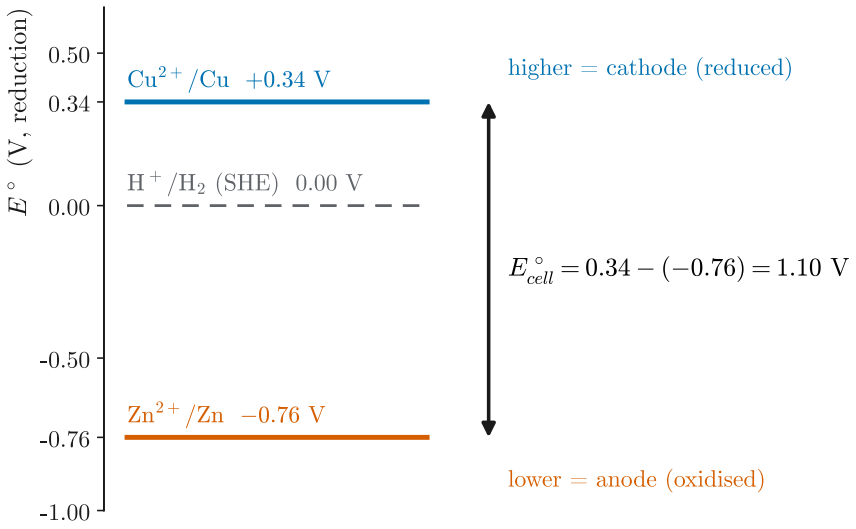


Figure G4.1. E°_{cell} as the gap between two electrodes on the potential scale. ESA ★★★

ADDED FOR CLARITY S001·9 · S001·10

How to read it. Place both reduction potentials on a vertical scale. The upper one is the cathode, the lower the anode, and E°_{cell} is the (positive) distance between them: $0.34 - (-0.76) = 1.10$ V for the Daniell cell.

5 The electrochemical series and its uses

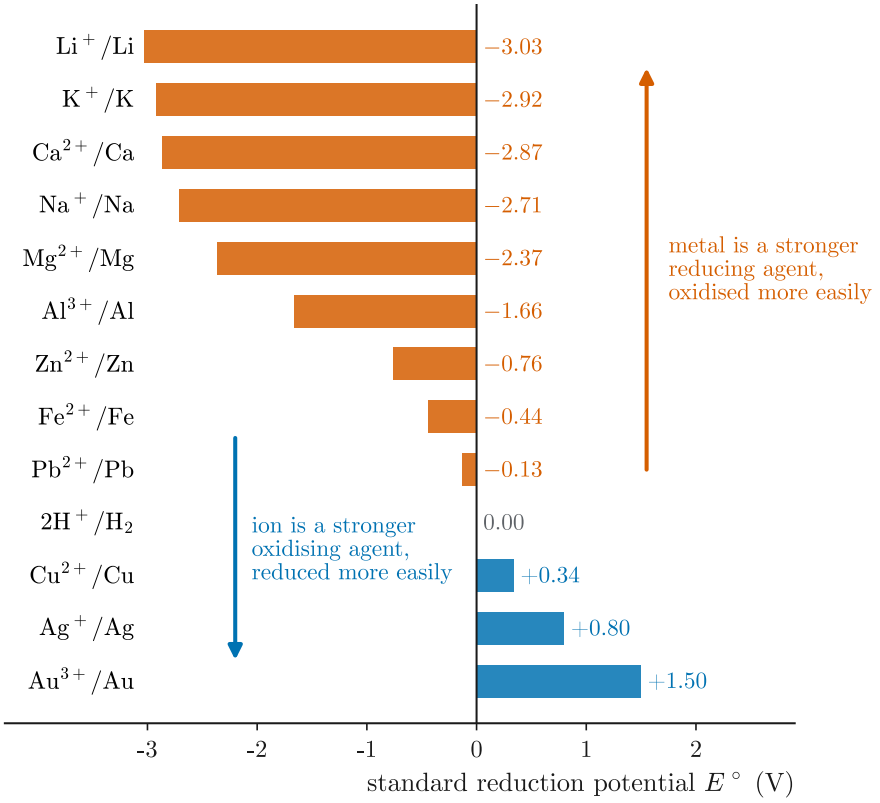
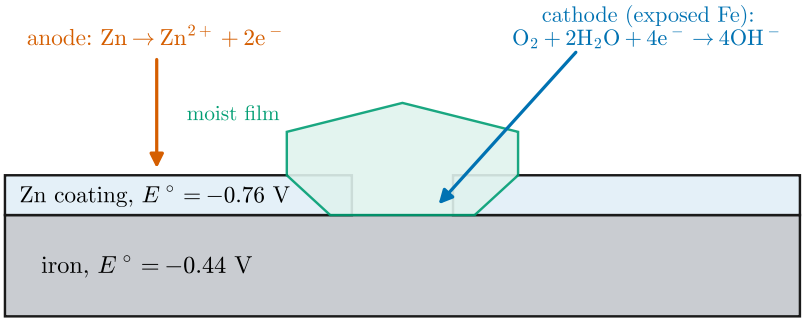


Figure G5.1. The electrochemical series (slide values). ESA ★★

S001·10

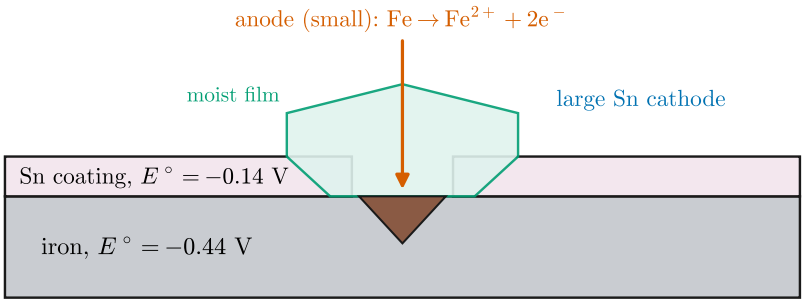
How to read it. Redrawn from the table on slide 001·10, with the same 13 couples and values. Bars to the left (negative E°) mark an oxidation tendency, bars to the right a reduction tendency. Upwards: the metal is the stronger reducing agent; downwards: the ion is the stronger oxidising agent. The slide lists Au at +1.50 V; ESA problems use 1.52 V.

(a) galvanised: Zn more active ⇒ Zn is the anode



iron is protected (sacrificial / cathodic protection) until the zinc is used up

(b) tinned: Fe more active ⇒ exposed Fe is the anode



small anode, large cathode: intense local corrosion (pitting) of the exposed iron

Figure G5.2. Scratched zinc and tin coatings on iron. ESA ★★

ADDED FOR CLARITY ESA Jul-23a 4d

How to read it. (a) Zn (**−0.76** V) is below Fe (**−0.44** V) in E°, so zinc is the anode and dissolves while exposed iron is a cathode: iron is protected. (b) Sn (**−0.14** V) is above Fe, so exposed iron is a small anode next to a large tin cathode and pits rapidly. Schematic, not to scale.